📦 Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

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📜 Project Overview

This project is designed to analyze and optimize the nationwide shipping operations of Nassau Candy Distributor. It leverages shipment and order data to evaluate delivery performance across factory-to-customer routes in the United States.

An interactive Streamlit dashboard is developed to provide actionable insights into logistics efficiency, helping improve delivery speed, reduce operational delays, and support data-driven decision-making.

🎯 Objective

The primary goal of this project is to enhance logistics efficiency by identifying:

- Fast and efficient shipping routes
- Delay-prone routes and regions
- Geographic bottlenecks in delivery networks
- Performance differences across shipping methods

❗ Problem Statement

Currently, the organization lacks clear visibility into end-to-end shipping performance across different routes and regions. This results in:

- Limited ability to proactively detect delays
- Inefficient route optimization
- Inconsistent evaluation of shipping methods
- Reactive rather than data-driven logistics decisions

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👥 Target Users

This solution is designed for:

- Logistics and Supply Chain Managers
- Operations Teams
- Business Executives
- Data Analytics and BI Teams

📊 Key Features

1. Overview Dashboard

Provides a high-level summary of logistics performance, including:

- Shipment KPIs and trends
- Delivery performance analysis
- Ship mode comparison
- Fastest and slowest route identification
- Delay distribution insights

2. Route Analysis

Focuses on performance evaluation of shipping routes:

- Identification of fastest and slowest routes
- Route ranking based on lead time
- Comparative route performance analysis

3. Geographic Analysis

Evaluates regional shipping efficiency:

- State-wise performance mapping
- Identification of delay hotspots
- Detection of geographic bottlenecks
- Regional classification of performance

4. Shipping Mode Analysis

Assesses performance of delivery methods:

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- Comparison of ship modes
- Lead time analysis
- Shipment distribution across modes
- Efficiency scoring

5. Factory Performance Map

Analyzes factory-level logistics efficiency:

- Factory performance monitoring
- Risk and bottleneck identification
- Geographic visualization of factory impact

6. Data Explorer

Enables detailed dataset exploration:

- Filtering and search capabilities
- Outlier and anomaly detection
- Data quality validation
- CSV export functionality

📂 Data Description

The dataset consists of nationwide shipment and order records, including:

- Order and shipment details
- Customer and product information
- Sales, cost, and profit metrics
- Geographic delivery data across U.S. states

Supporting Data

- 🏭 Factory location coordinates (latitude & longitude)
- 🔗 Product-to-factory mapping for route analysis

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📈 Key Performance Metrics

Metric	Description
Avg Lead Time	Average delivery duration per shipment
Total Shipments	Total number of shipment records
Delay Rate	Percentage of delayed deliveries
Efficiency Score	Delivery speed performance indicator
Bottleneck Score	Combined delay and lead time impact
Factory Risk Level	Classification of factory performance

🧭 Application Workflow

1. Launch the Streamlit dashboard
2. Dataset loads automatically
3. Apply filters (region, state, ship mode, route)
4. Explore analytics dashboards:
    - Overview
    - Route Analysis
    - Geographic Analysis
    - Shipping Mode Analysis
    - Factory Performance
5. Use Data Explorer for detailed inspection and export

🛠️ Technologies Used

- Python (Core development)
- Streamlit (Interactive dashboard)
- Pandas (Data processing)
- Plotly Express & Mapbox (Visualization)
- NumPy (Numerical analysis)

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- CSV-based dataset storage

⚠️ Limitations

- Performance may reduce with large datasets
- Geographic analysis limited to U.S. regions
- Some factory metrics are simulated or estimated
- Accuracy depends on completeness of shipment data
- Dashboard requires manual refresh after data updates

🚀 Future Enhancements

- Predictive analytics for shipment delays using AI/ML
- Real-time shipment tracking integration
- Expansion to global logistics networks
- Cloud deployment for web access
- Automated alert system for delays and bottlenecks
- Enhanced forecasting and trend modeling

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